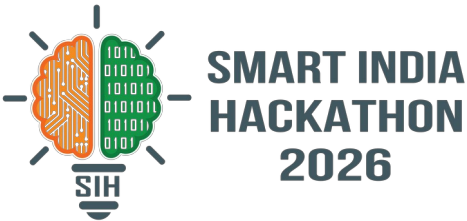


SMART INDIA HACKATHON 2026



RASTA AI

Route Intelligence for Essential-Supply Logistics in North Eastern India

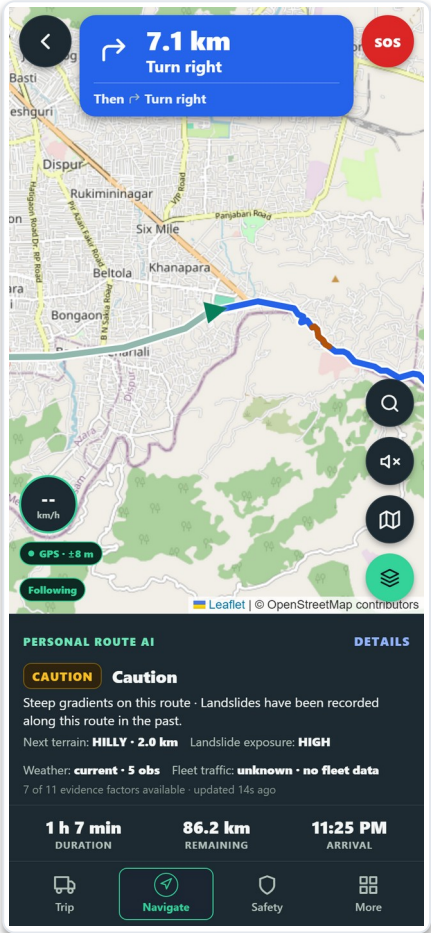
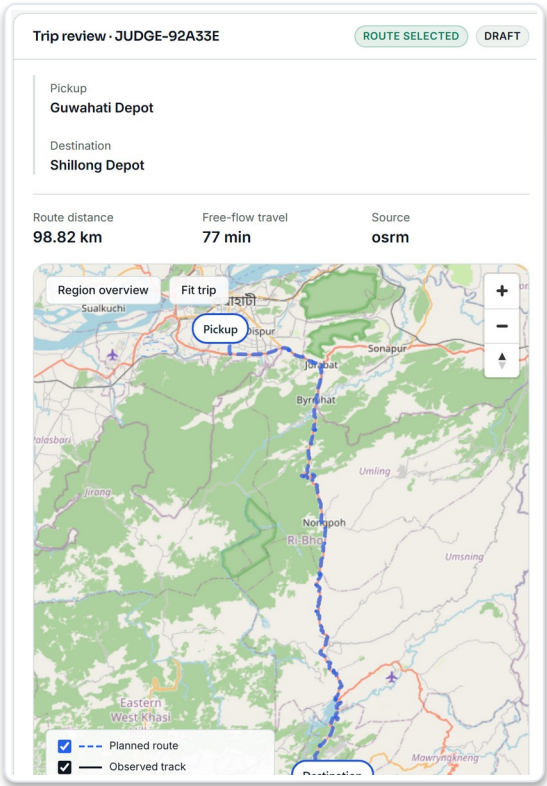
SHORTEST IS NOT ALWAYS USABLE.

“Can this essential-supply truck reliably use this corridor now?”

PROBLEM STATEMENT ID	SIH26002
THEME	Smart Automation
PS CATEGORY	Software
TEAM ID	17 (internal group no.)
TEAM NAME	NER-AI LOGISTICS

AI-Based Smart Logistics and Accessibility Intelligence Platform for North Eastern Region (NER)

LIVE WORKING SYSTEM



Manager route review · Driver navigation — real screens, real road

PROPOSED SOLUTION

Not the shortest road. **The road that is usable now.**

Monsoon · landslides · floods · official warnings · single-lane hill roads · no signal. **A shortest-path router still sends the truck.**

A NORMAL ROUTER

Origin → Shortest / fastest road → Truck

No governance. No answer when data is missing.

RASTA AI

Terrain

Weather

Landslide exposure

Flood context

Official warnings

Fleet traffic

Evidence freshness

CONTINUE

CAUTION

HOLD

REROUTE

Manager governance → driver navigation, 23 languages, offline package

UNKNOWN ≠ SAFE

Missing or stale evidence is marked UNKNOWN — never safe — and goes to an authorised reviewer.

MANAGER · CHECK CONDITIONS

REQUIRES REVIEW

Hazard evidence is incomplete or elevated. An authorised reviewer must review this route before selection. Refresh conditions after review.

Landslide data is not available - no source is connected. · Steep gradients on this route · Landslides have been recorded along this route in the past. · The landslide inventory ends in 2017 - recent years are not covered. · River levels near this route are close to their recent average. · No official disaster alert names a district on this route right now. · No fleet truck has driven this road in the last fifteen minutes, so traffic is unknown.

Unavailable: Fuel range, Landslide, Road quality, Truck restrictions

Evidence coverage · Weather AVAILABLE · Terrain AVAILABLE · Warnings AVAILABLE · Traffic UNKNOWN

TERRAIN

0.5 KM STEEP

49-1436 m · 1975 m climb · steepest 10.3%

flat 61.8 km · rolling 29.6 km · hilly 7.0 km · steep 0.5 km

Copernicus DEM GLO-90 via Open-Meteo + SRTM 30 m via OpenTopoData · 100% of 198 samples answered

LANDSLIDE HISTORY

HIGH EXPOSURE

22 recorded landslides within 5 km of the road, nearest 0.4 km

6 more nearby placed too imprecisely to count · HIGH is three or more; MODERATE is one or two

NASA Global Landslide Catalog · inventory 2007-2017 — aged, recent years not covered

RIVER LEVELS

NORMAL

Discharge at 5 river cells along the corridor, highest 1.0x its own 30-day mean · GloFAS via Open-Meteo, 2026-09-14. A level, not a flood forecast and not a road-closure claim.

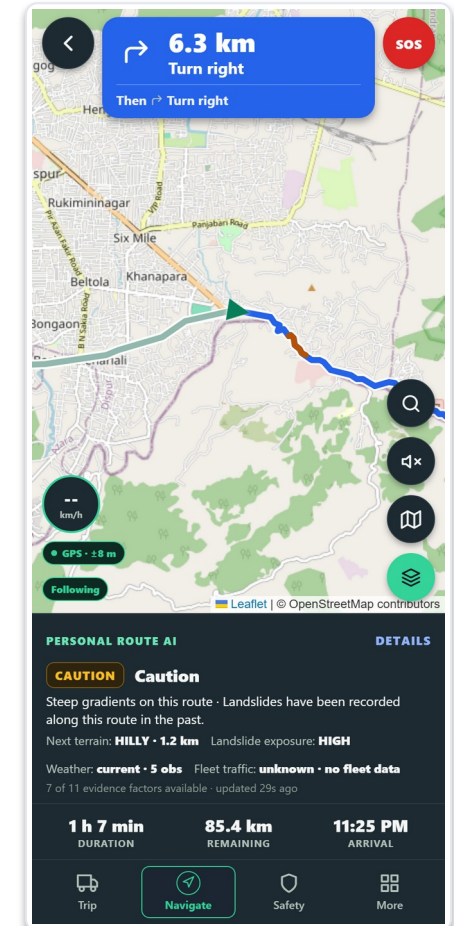
OFFICIAL ALERTS

NONE ON CORRIDOR

NDMA SACHET CAP feed, 99 alerts nationwide at 10:13:54 PM · placed by district name (Dispur, East Khasi Hills, Kamrup Metropolitan, Myllem, Ri-Bhoi, Umiling, Umsning) · 4 more active elsewhere in the corridor states, not placeable on this road.

FLEET TRAFFIC

UNKNOWN



Manager evidence · Driver route AI — the same decision, both ends

VERIFIED DATA

WEATHER

Open-Meteo · MET
Norway

TERRAIN

Copernicus DEM ·
OpenTopoData

WARNINGS

NDMA SACHET
(CAP)

FLOOD

GloFAS

LANDSLIDE

NASA GLC 2007–
17

TRAFFIC

fleet GPS fixes

ROUTE-SPECIFIC EVIDENCE · FastAPI backend

every factor sampled along the selected road · PostgreSQL + PostGIS (Supabase) · OSRM routes, turn steps

DETERMINISTIC SAFETY POLICY

11 factors · reason codes · UNKNOWN ≠ SAFE · reviewer authorisation when hazard data is missing

CONTINUE

CAUTION

HOLD

REROUTE

MANAGER REVIEW

DRIVER NAV

60 s WATCH

REASSESS

Re-scored every 60 s on the road ahead. A reroute is proposed to the manager, never applied silently.

DETERMINISTIC SAFETY ENGINE

makes every operational decision

GEMINI / OPENROUTER (LLM)

explains the decision in the driver's language — never decides

EXPERIMENTAL ML

research channel only; held outside routing by the safety gate

STACK · CLIENTS · GOVERNANCE

CLIENTS Manager console (React + TS) · Driver app (React Native / Expo, Android) · one login, server-decided role, offline package**STACK** FastAPI · PostgreSQL / PostGIS (Supabase) · React · React Native / Expo · OSRM · OpenStreetMap · Gemini

GOVERNANCE CHAIN



Diagnostics reports every provider's real state — healthy, degraded, fallback active or unknown — so a judge can see exactly what a decision rested on.

Not a concept. A working end-to-end prototype.

CERTIFIED LIFECYCLE

PLAN

ACCEPT

OFF-ROUTE

REVIEW

TRUCK VERIFY

MANAGER REROUTE
APPROVAL

DISPATCH

NAVIGATE

DELIVERY

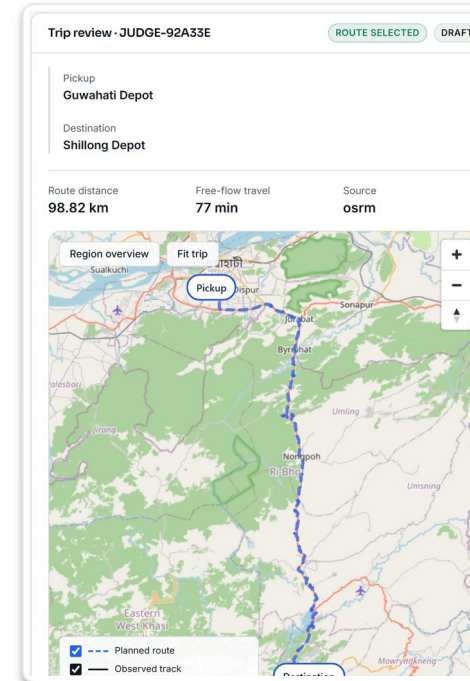
12/12 physical judge-flow steps**9/9** physical role-flow steps**1138+** backend · **621+** driver · **170+** manager tests

PHYSICAL ANDROID · CERTIFIED

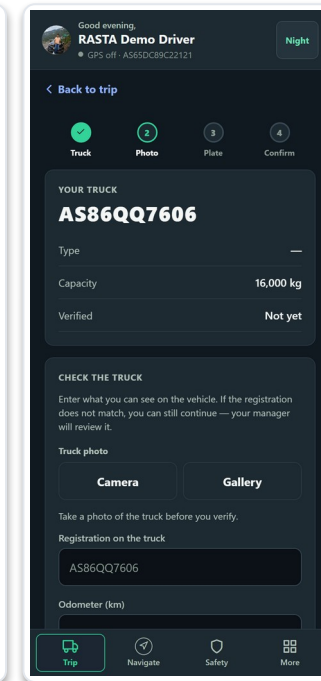
Canonical demo Guwahati → Shillong · ≈ 98.8 km real route

KNOWN LIMITS → MITIGATION Cold start → warm /health first · rate limits → cached evidence + explicit
UNKNOWN · push needs Firebase → in-app alerts work · experimental ML → outside routing

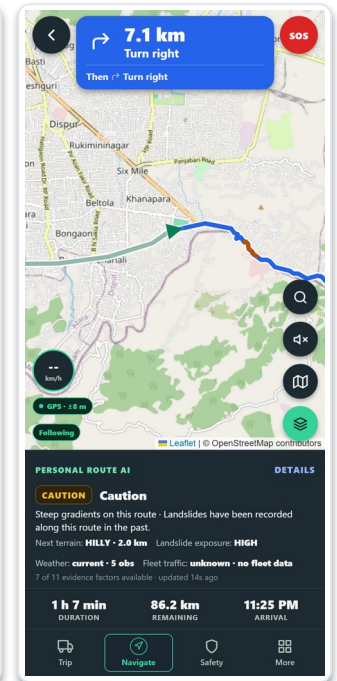
REAL SCREENS FROM THE CERTIFIED RUN



Manager · route review



Driver · truck check



Driver · navigation

“Can this truck reliably use this corridor now?”

FOOD · MEDICINE · FUEL · RELIEF SUPPLIES

essential logistics for hill communities, decided on evidence

FLEET MANAGERS

Usability and uncertainty visible before dispatch. Auditable.

DRIVERS

Danger context, emergency support, 23 languages.

REMOTE COMMUNITIES

Resilient access to food, medicine, fuel, relief.

GOVERNMENT / OPS

Auditable evidence and disruption per corridor.

23 selectable languages, explicit fallback status

11 route factors in every decision

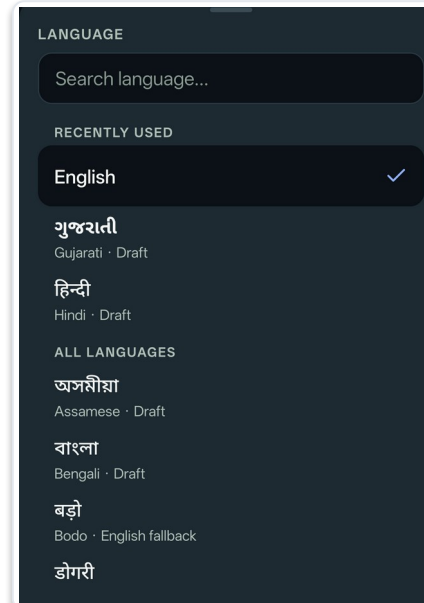
60 s route-ahead monitor while driving

12/12 judge-flow steps certified on a physical phone

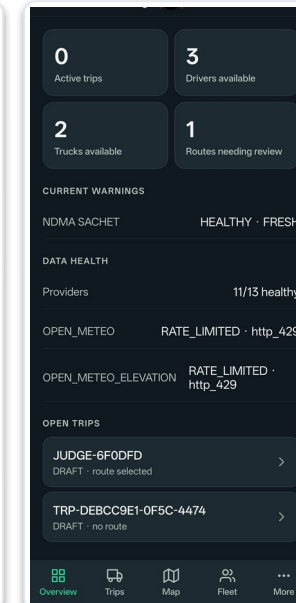
100 % of reroutes reviewed by a human

0 unknown factors counted as safe

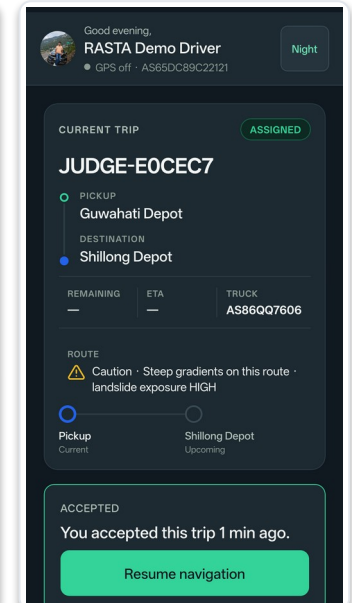
REAL SCREENS · APK 1.0.18



23 languages



Manager · mobile



Driver · assigned trip

HUMAN GOVERNANCE — AI HAS NO UNCONTROLLED AUTHORITY



EXPERIMENTAL LANDSLIDE RESEARCH

95 %

RECALL

NER geographic holdout · 154 events · no NER row in training

SAFETY GATE
NOT DEPLOYED

FPR 50 % too high for production deployment

TRAIN

HELD-OUT NER TEST

USEFUL SIGNAL FOUND

FALSE POSITIVES TOO HIGH

SAFETY GATE

NOT DEPLOYED

VALIDATION BEFORE AUTOMATION. Recall was not enough. Held-out testing exposed the false-positive rate. The gate keeps it out of routing.

Precision 0.32 · F1 0.48 · PR-AUC 0.55 · ROC-AUC 0.83 · Brier 0.12 — calibrated logistic regression, NASA GLC + ERA5-Land rain + DEM slope

RESEARCH / EVIDENCE SOURCES BEHIND THE ROUTE DECISION

ROUTING / PLACES

OSRM · OpenStreetMap · Nominatim

Road geometry, alternatives, turn steps.

WEATHER / TERRAIN

Open-Meteo · MET Norway · Copernicus DEM · OpenTopoData

Forecast along the road; climb, gradient, steep segments.

OFFICIAL / FLOOD

NDMA SACHET (CAP) · Copernicus GloFAS

Official warnings with freshness; discharge vs 30-day mean.

LANDSLIDE HISTORY

NASA Global Landslide Catalog

Historical exposure, 2007–2017, ≤5 km accuracy.

AI ASSISTANCE

Google Gemini

Explains in the driver's language. Never safety authority.

PROBLEM STATEMENT

MDoNER · SIH26002

Smart logistics and accessibility intelligence for the NER.

REFERENCES gpm.nasa.gov/landslides · sachet.ndma.gov.in · global-flood.emergency.copernicus.eu · open-meteo.com · api.met.no · opentopodata.org · project-osrm.org · openstreetmap.org · github.com/nxtlucifer/ner-ai-logistics